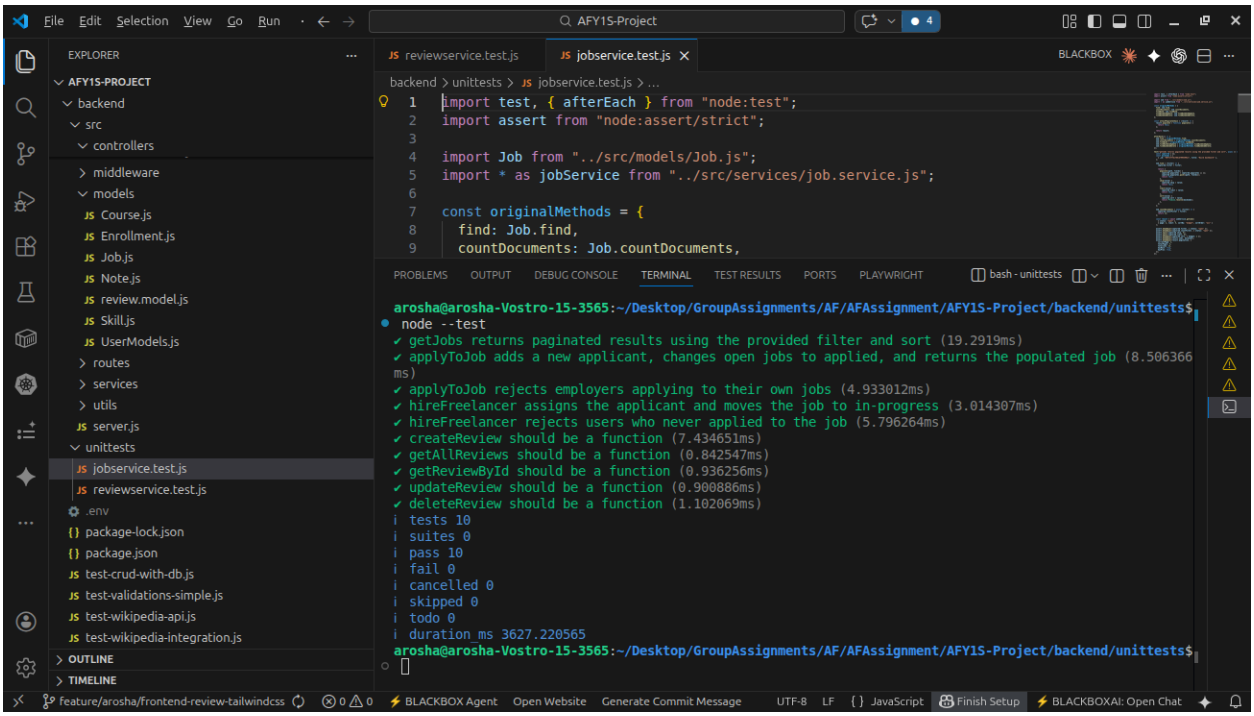
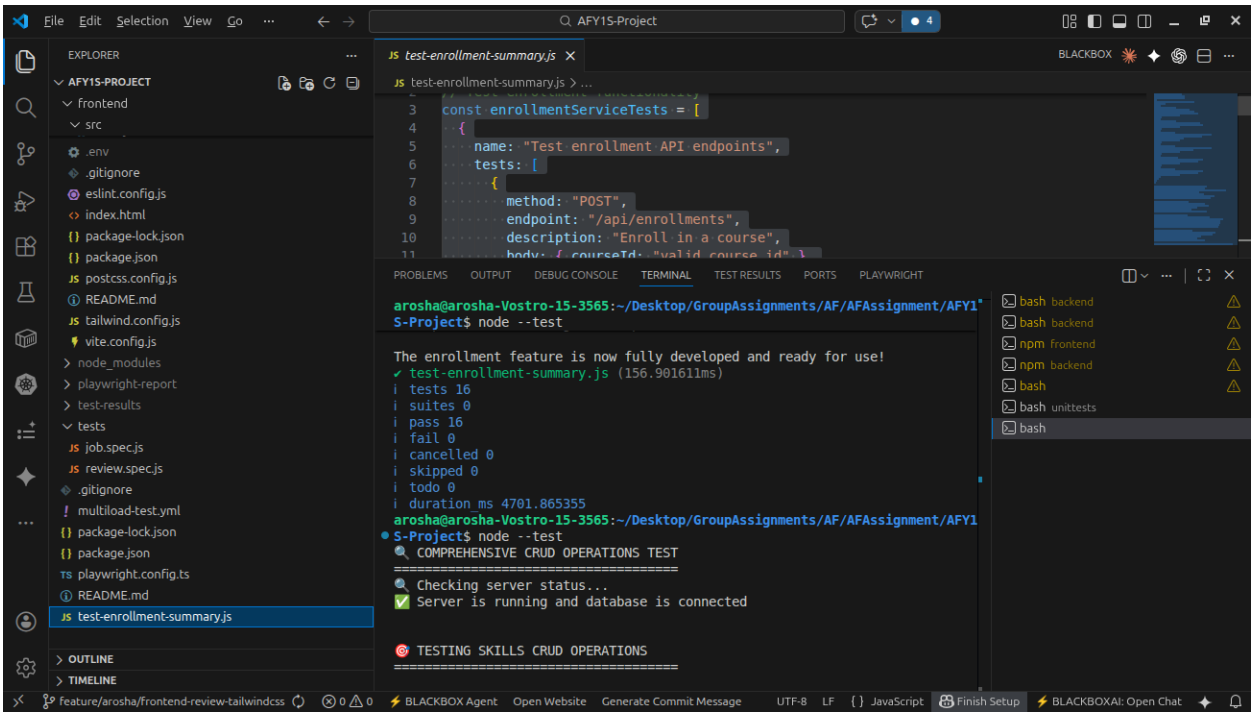
Unit Testing:

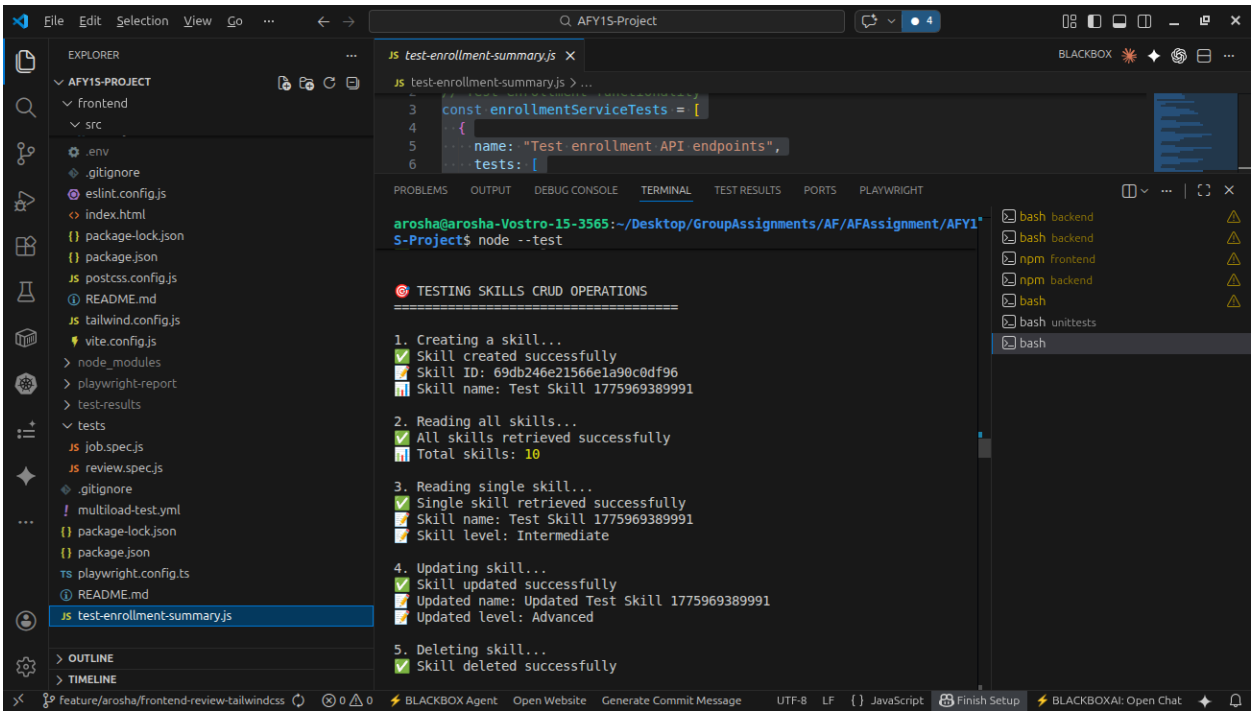




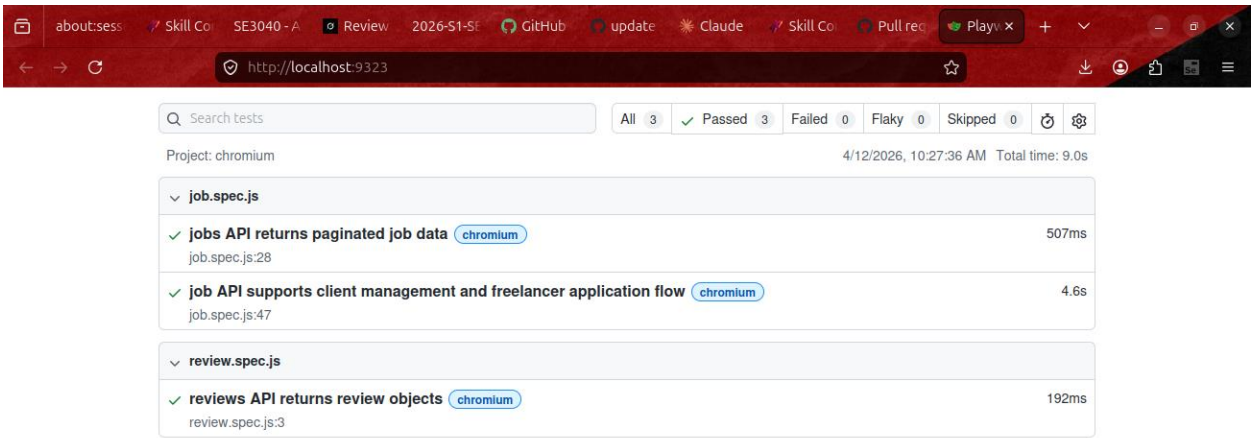
```
🎯 SKILLS VALIDATION TESTS
------------------------------------

1. Testing name validation rules...
✅ empty string: Invalid (Expected)
✅ too short (1 char): Invalid (Expected)
✅ valid name: Valid (Expected)
✅ too long (101 chars): Invalid (Expected)
✅ valid with special chars: Valid (Expected)
✅ invalid special chars: Invalid (Expected)

2. Testing level validation rules...
✅ valid beginner: Valid (Expected)
✅ valid intermediate: Valid (Expected)
✅ valid advanced: Valid (Expected)
✅ invalid level: Invalid (Expected)
✅ empty level: Invalid (Expected)

📚 COURSES VALIDATION TESTS
------------------------------------

1. Testing title validation rules...
❌ too short (5 chars): Valid (Expected Invalid)
✅ valid title: Valid (Expected)
✅ too long (201 chars): Invalid (Expected)
✅ valid with punctuation: Valid (Expected)

2. Testing duration validation rules...
✅ zero duration: Invalid (Expected)
✅ minimum valid duration: Valid (Expected)
✅ valid duration: Valid (Expected)
✅ maximum valid duration: Valid (Expected)
✅ exceeds maximum: Invalid (Expected)
✅ negative duration: Invalid (Expected)

3. Testing MongoDB ObjectId validation...
✅ valid ObjectId: Valid (Expected)
✅ invalid format: Invalid (Expected)
✅ empty string: Invalid (Expected)
✅ too short: Invalid (Expected)

👥 ENROLLMENTS VALIDATION TESTS
------------------------------------

1. Testing progress validation rules...
```

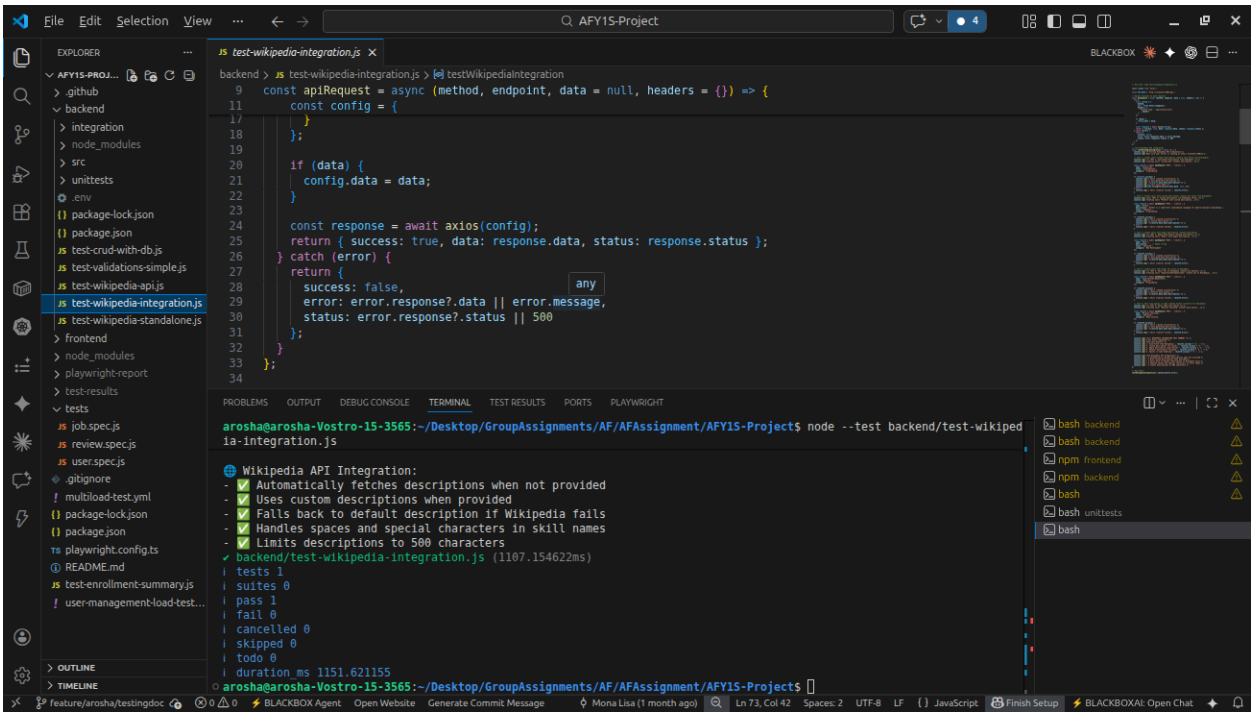
```
const enrollmentServiceTests = [
  {
    name: "Test enrollment API endpoints",
    tests: [
```

```
arosha@arosha-Vostro-15-3565:~/Desktop/GroupAssignments/AF/AFAssignment/AFY1S-Project$ node --test
📚 TESTING COURSES CRUD OPERATIONS
==================================

0. Creating a skill for course reference...
✅ Reference skill created with ID: 69db247021566e1a90c0dfa6

1. Creating a course...
✅ Course created successfully
📝 Course ID: 69db247021566e1a90c0dfaa
📊 Course title: Test Course 1775969391109

2. Reading all courses...
✅ All courses retrieved successfully
📊 Total courses: 7

3. Reading single course...
✅ Single course retrieved successfully
📝 Course title: Test Course 1775969391109
🔗 Skill populated: Course Test Skill 1775969391109

4. Updating course...
✅ Course updated successfully
📝 Updated title: Updated Test Course 1775969391109
📝 Updated duration: 60

5. Deleting course...
✅ Course deleted successfully
```



Integration testing





Load Testing

```
arosha@arosha-Vostro-15-3565:~/Desktop/GroupAssignments/AF/AFAssignment/AFY1S-Project$ npx artillery run multiload-test
.yml
  p99: ..................................................................... 31266.3


All VUs finished. Total time: 2 minutes, 30 seconds

--------------------------------
Summary report @ 10:38:16(+0530)
--------------------------------

errors.ETIMEDOUT: ........................................................ 334
http.codes.200: .......................................................... 7224
http.downloaded_bytes: ................................................... 17585042
http.request_rate: ....................................................... 51/sec
http.requests: ........................................................... 7742
http.response_time:
  min: ................................................................... 0
  max: ................................................................... 5529
  mean: .................................................................. 1222.2
  median: ................................................................ 699.4
  p95: ................................................................... 3984.7
  p99: ................................................................... 4867
http.response_time.2xx:
  min: ................................................................... 0
  max: ................................................................... 5529
  mean: .................................................................. 1222.2
  median: ................................................................ 699.4
  p95: ................................................................... 3984.7
  p99: ................................................................... 4867
http.responses: .......................................................... 7224
vusers.completed: ........................................................ 506
vusers.created: .......................................................... 840
vusers.created_by_name.User opens frontend and browses system: ........... 840
vusers.failed: ........................................................... 334
vusers.session_length:
  min: ................................................................... 19551.9
  max: ................................................................... 43152.3
  mean: .................................................................. 29905.7
  median: ................................................................ 30040.3
  p95: ................................................................... 40550.5
  p99: ................................................................... 42205.5
arosha@arosha-Vostro-15-3565:~/Desktop/GroupAssignments/AF/AFAssignment/AFY1S-Project$
```